

Space Computing Use Case

AI OS 0xMeridian + Magellano Inference Engine
Quantum-Assisted Orbital Edge AI with Federated Constellation Governance

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"The Constellation does not ask for trust. It asks for proofs. In space, proofs are all you have." — ADR-030

1. Executive Summary

UC03 defines the complete architecture for deploying AI OS 0xMeridian and the Magellano inference engine in orbital edge computing environments. Three operational modes are fully specified: standalone satellite node, disaster-recovery failover, and n-node federated constellation. Every design decision is traced to a physics boundary, a complexity class, or a hardware constraint.

The document integrates TDD v5.1 (terrestrial AI OS) with TDD v5.2 (Space Computing Extension), covering quantum-assisted search (ADR-032), space-grade self-learning (ADR-033), federated learning privacy bounds (ADR-033bis), concept drift detection (ADR-033ter), federated constellation governance (ADR-030), and QKD inter-satellite recovery.

1.1 Three Core Principles

Principle	Technical Implementation	Competitive Advantage
Sovereignty	Zero ground-cloud dependency; every node fully autonomous	Operational during multi-hour/day communication blackouts
Provability	ZK-proofs (192 B Groth16) for every critical operation	Verification without trust — essential for multi-nation constellations
Adaptability	QLoRA self-learning with SEU-resilient checkpoints	Continuous autonomous improvement without human intervention

1.2 Key Performance Indicators

Metric	Value	Source / Notes
Magellano inference latency	50–100 ms on-orbit	ADR-031; Apple Silicon Metal FP16 baseline
Magellano power draw	~15 W (NF4 quantised)	ADR-031; leaves 25 W margin in sunlit mode
Grover speedup ($N=10^{26}$)	1000× vs $O(N)$ classical	ADR-032 Stage 1 simulation; 50 s → 50 ms
QAOA task allocation ($p=4$)	280 ms, 95% of ILP optimal	ADR-032; M4 NPU; 10 nodes, 50 tasks
NISQ quantum advantage crossover	$N \approx 200\text{--}500$ nodes	ADR-032; where QAOA beats classical heuristics
ZK-proof size (Groth16)	192 bytes	ADR-013
ZK verification time	<2 ms @ 10 M proofs/s	ADR-013 benchmarks
Raft re-election	<500 ms	ADR-004 + ADR-030

Metric	Value	Source / Notes
Node recovery at T+5 s	80% coverage restored	ADR-030 constellation
Federated round overhead	<120 s / 90-min orbit	ADR-033 Phase A–D
SMPC overhead vs FedAvg	<5% latency	ADR-033bis
SEU circuit fidelity (LEO)	>99.5%	ADR-032 three-layer protection
Disaster recovery freeze→safe	<5 min	ADR-007 + ADR-033
Full recovery after link restore	<30 min	ADR-033 Mode 2

Zero-regression guarantee: every quantum or advanced component has a classical fallback (FAISS + simulated annealing). Quantum paths are optimisation, not dependency.

2. Strategic Context & Market Landscape

2.1 Global Space Computing Market (2025–2030)

The space computing market is entering a critical transition: from ground-controlled edge nodes to truly autonomous orbital AI. Key investments shaping the landscape:

- **USA:** NASA "Moon to Mars" AI initiative (\$2.3 B FY2026); Space Force "Orbital Prime" for debris management.
- **EU:** ESA "Boost!" programme (€350 M); EuroHPC-JU quantum-space call (€80 M); ESA ASCEND orbital data-centre demo (2026).
- **China:** "Guowang" constellation (13,000 satellites planned); Micius-2 quantum satellite network.
- **Commercial:** Starlink (10,000+ satellites); Amazon Kuiper (3,236); Planet Labs (AI-driven Earth observation); IonQ/Capella Space QKD (July 2025).

Market Gap: No player currently offers autonomous AI with verifiable self-learning. Existing solutions (D-Orbit, Loft Orbital, Spire) provide edge computing but require continuous ground control. 0xMeridian closes this gap with three differentiators: Autonomous QLoRA (learning without ground truth), ZK-Verified Operations (auditability for government clients), and Quantum-Ready architecture (upgrade path without redesign).

2.2 Competitive Landscape

Feature	0xMeridian	AWS Snowcone	Azure Orbital	D-Orbit ION	Alibaba Qwen3+ASIC
Kernel	Rust (zero GC)	Linux	Linux	Linux	Custom ASIC
AI Model	Magellano 3.3B NF4	Container generic	Cloud-dependent	Limited	4-bit MoE
Power draw	~15 W active	N/A	N/A	N/A	~50 W active
Self-Learning	QLoRA + SEU resilience	No	No	No	No
Federation	Native ZK-proofs	Ground-dependent	Ground-dependent	Manual	No
SEU protection	CRC-32 + MRAM + QEC	None	None	None	None documented
Quantum-Ready	Three-stage path	No	No	No	No
Recovery	<5 min autonomous	Ground-dependent	Ground-dependent	Manual	No
Open Source	Yes (github.com/Gyros4me)	No	No	No	No

3. Physical Constraints of Orbital Computing

Space computing is not "edge computing with latency." It is a fundamentally different operational domain. Five hard physical constraints invalidate every assumption of terrestrial AI architecture.

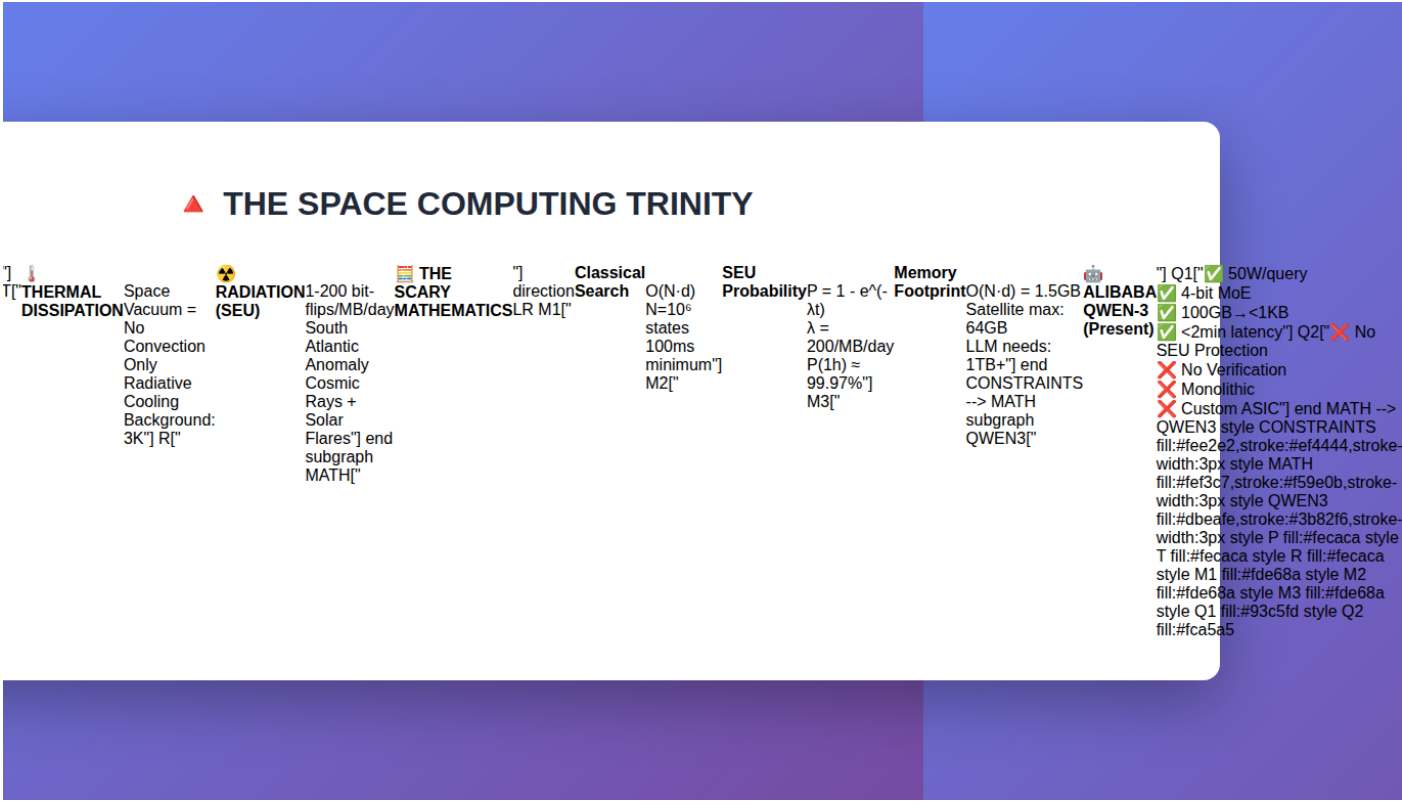


Figure 1 — The Space Computing Trinity: Power, Radiation, Connectivity

3.1 Environmental Parameters by Orbit

Parameter	LEO (400–2000 km)	MEO (8,000–20,000 km)	GEO (35,786 km)	Deep Space (>1 AU)
Orbital period	90–127 min	~6 hours	24 hours (sync)	N/A
Ground contact	8–15 min/pass	2–8 hours/day	Continuous	Hours (DSN)

Parameter	LEO (400–2000 km)	MEO (8,000–20,000 km)	GEO (35,786 km)	Deep Space (>1 AU)
Latency (RTT)	2.6–13.4 ms	53–133 ms	238 ms	6–44 min
Link budget	2–50 Mbps (X-band)	1–10 Mbps	150 Mbps (Ka)	2–8 Kbps (X-band)
Radiation (TID/yr)	1–10 krad	10–100 krad	5–50 krad	>100 krad
SEU rate	1–10 bit-flips/MB/day	5–20 bit-flips/MB/day	10–50 bit-flips/MB/day	50–200 bit-flips/MB/day
Thermal cycling	+120°C / -180°C	±100°C	±150°C	±200°C
Power budget	50–200 W	100–300 W	100–500 W	50–150 W (RTG)

3.2 Single Event Upsets (SEU): The Silent Killer

High-energy protons (solar proton events) and cosmic rays (heavy ions) deposit charge in DRAM/SRAM memory nodes, causing bit-flips. In a Q8 quantised model (8-bit), a flip of bit 7 (MSB of mantissa) changes a weight from 1.0 to 1.9375 — a 94% error. In NF4 (4-bit), the outlier-detection property provides inherent resilience: a bit-3 flip generates a ± 4 -sigma outlier that is immediately flagged.

0xMeridian mitigation stack: (1) NF4 quantisation with outlier detection; (2) CRC-32 per 64-weight block with 10 s scrub cycle; (3) MRAM checkpoints every 100 training steps; (4) Three-layer quantum circuit protection (physical shielding + Steane $[[7,1,3]]$ QECC + majority vote).

3.3 Complexity Class Boundary

The critical architectural decision is not about frameworks — it is about complexity class. Transformer attention is $O(n^2)$ in sequence length. At 40 W power envelope, this is mathematically impossible beyond ~512 tokens. Mamba state-space models are $O(n)$ — this is the boundary that makes orbital inference tractable. For search: classical brute force $O(N)$ over $N=10^6$ debris objects = 50 s. Grover $O(\sqrt{N})$ simulated on NPU = 50 ms. For NP-hard task allocation, QAOA achieves 95% of optimal ILP in 280 ms.

4. System Architecture

4.1 Magellano Inference Engine

Magellano is a 3.3 B parameter Mamba-MoE architecture designed for constrained edge deployment. 30 Mamba layers (linear recurrence, $O(n)$) combined with 9 Mixture-of-Experts routing layers for task specialisation without full-parameter activation.

Parameter	Value	Orbital Relevance
Architecture	30 Mamba layers + 9 MoE routing	$O(n)$ complexity fits 40 W envelope
Parameters	3.3 B total; ~0.5% active/token	Selective activation minimises power
Quantisation	NF4 — 6.6 GB → 1.7 GB	Fits rad-hardened LPDDR4 budget
Inference latency	50–100 ms (Metal FP16)	8.81× vs CPU baseline
Power draw	~15 W active inference	Leaves 25 W margin in sunlit mode
Hot-swap	Zero-downtime adapter swap	No orbit interruption during update
Self-learning	QLoRA rank-16, 7 modules	~35 MB adapter delta per orbit round

4.2 Kernel Stack — Three-Language Polyglot

Each language is chosen for deterministic correctness at its layer. Rust handles the kernel (zero GC, no unpredictable pauses). OCaml + Idris 2 handles orchestration (algebraic types, formal contract verification). Swift + Metal handles inference (CoreML/NPU acceleration, 8.81× speedup).

Layer	Language	Component	Key Property
Kernel	Rust	Resource Manager, Circuit Breaker, ZK Prover	Zero GC, deterministic
Orchestration	OCaml + Idris 2	Task Scheduler, SDL contracts, Agent Swarm	Formal verification
Inference	Swift + Metal	Magellano, CoreML, NPU acceleration	8.81× vs CPU baseline
Bus 1	gRPC/Protobuf	Agent ↔ Kernel commands	<10 ms, mTLS + JWT
Bus 2	NATS JetStream	Agent ↔ Agent events	10–200 ms, pub/sub fan-out
Bus 3	Shared Memory	Kernel ↔ Inference hot path	<1 ms, zero-copy DMA

Layer	Language	Component	Key Property
Bus 4	WireGuard + DTN	Inter-satellite ZK-SLA proofs	192 B proofs, DTN custody

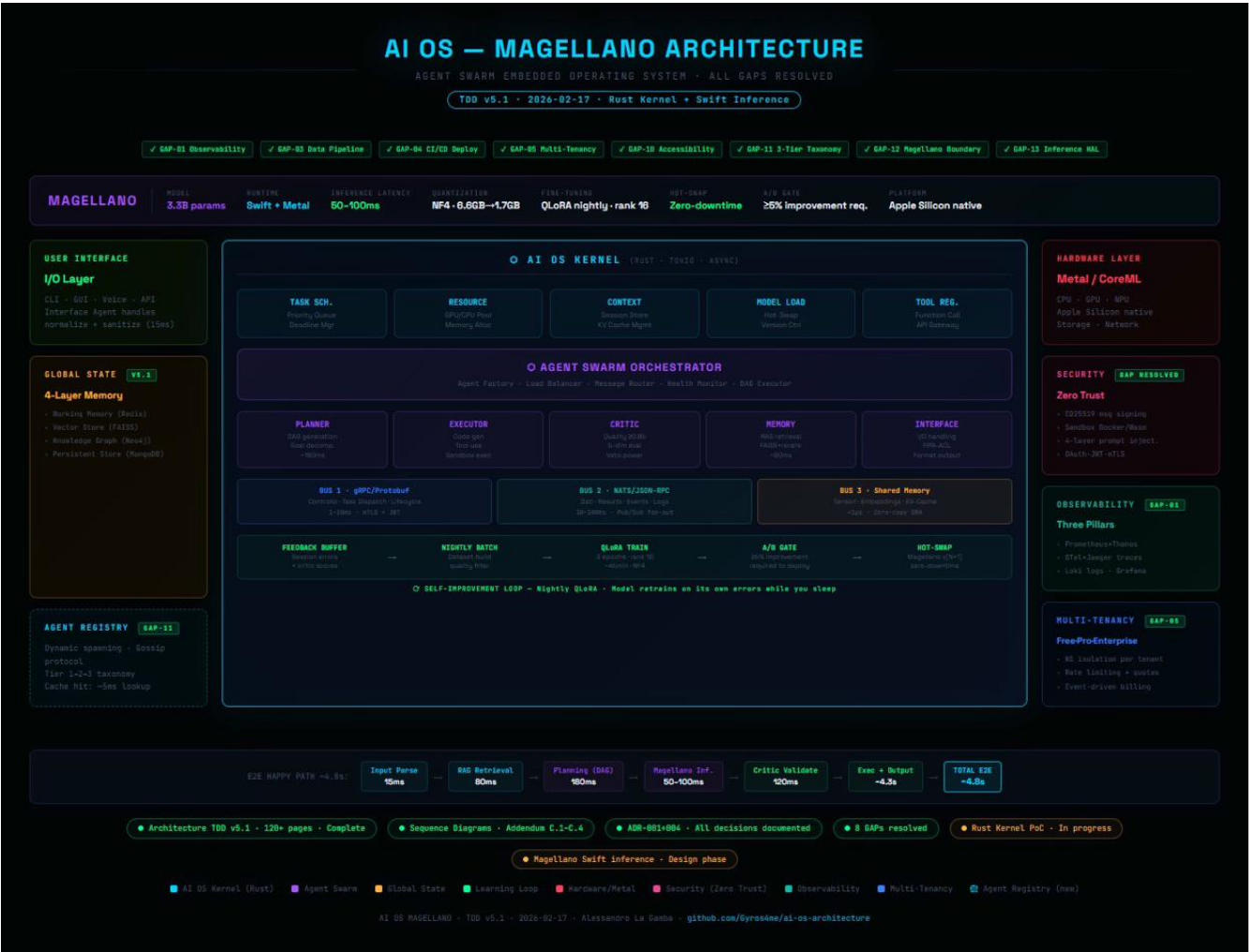


Figure 2 — AI OS 0xMeridian TOD v5.1: Full Architecture Manifesto

5. Three Orbital Operational Modes

0xMeridian operates in three mutually exclusive modes with automatic kernel-managed transitions based on power, radiation, and connectivity state.

Mode	Trigger	Characteristics	Use Case
STANDALONE	Default	Full autonomy, local learning, no uplink dependency	LEO routine, deep space
DISASTER RECOVERY	SEU storm; link loss >3 windows; battery <20%	Freeze training, integrity audit, selective rollback, degraded inference	Solar proton event, debris collision
CONSTELLATION MESH	≥2 nodes reachable via DTN	Federated learning, SMPC gradient aggregation, ZK-SLA proofs	Walker constellation, formation flying

5.1 Mode 1 — AUTONOMOUS-QLoRA (Standalone)

Single satellite, no peers. LEO orbital cycle (90 min):

Time	Phase	Actions
T+0 min	SUNLIT — Active	Magellano inference nominal; feedback buffer accumulation; CRC scrubber 10 s cycle
T+44 min	Pre-Eclipse Checkpoint	Serialise LoRA adapter + optimiser state to MRAM (<30 s for adapter delta ~400 MB)
T+45 min	Eclipse — Power-Constrained	Micro-epoch training (1/4 normal); gradient clip 0.5; batch=2; SEU guard per batch
T+90 min	Eclipse Exit — Resume	Verify CRC chain; resume or rollback; emit ModelVersionManifest with orbit_id
T+92 min	Contact Window (if available)	Upload delta adapter ~50 MB compressed; DTN Bundle Protocol RFC 5050

5.2 Mode 2 — SAFE-QLoRA (Disaster Recovery)

Triggered by: uplink silence >3 contact windows (270 min LEO); SEU rate >10 events/hour; battery <20%; or peer SWIM gossip declaring this node SUSPECTED.

Phase	Time	Action	Output
FREEZE	T+0 s	Halt training, flush buffer, broadcast SAFE_MODE	FIPA-ACL INFORM
INTEGRITY AUDIT	T+5 s	Full CRC scan (3.3 B model = 6.6 GB @ 100 MB/s ≈ 66 s)	ZK-proof of audit state
SELECTIVE ROLLBACK	T+2 min	Corrupt >5%: full rollback N-1; 1–5%: selective restore; <1%: inline correction	Model integrity report
DEGRADED INFERENCE	T+5 min	Inference only; ONNX fallback if integrity <95%; telemetry every 60 s	Telemetry stream

Phase	Time	Action	Output
RECOVERY	Link restored	Upload audit + SEU log; download validated checkpoint; gradual re-enable	Full ops restored

KPI: Freeze-to-safe-inference <5 min; integrity audit <2 min; full operational recovery <30 min after link restore.

5.3 Mode 3 — FEDERATED-QLoRA (Constellation)

90-minute LEO orbit defines the federated learning round. All gradient transmission uses SMPC pairwise masking (ADR-033bis).

Phase	Time	Actions
A: Local Training	T+0..75%	AUTONOMOUS-QLoRA on local data; produce G_i (~35 MB); sign Groth16 ZK-proof {node_id, round_id, num_samples, final_loss}
B: Gossip Broadcast	T+75..80%	Broadcast SHA256(G_i) via SWIM gossip over Bus-4 (DTN bundles); O(log N) convergence ~30 s for 10 nodes
C: Gradient Aggregation	T+80..85%	Raft leader: FedAvg with reputation weighting; SMPC masking; DP noise eps=1.0, delta=1e-5; ZK-SLA proof
D: Hot-Swap	T+85..90%	Raft 2/3 quorum on G_agg; atomic pointer swap <3 s; rollback if Critic score drops >5%

DTN Latency Budget (10-node LEO, X-band):

Component	Time
Gossip convergence (RTT)	30–45 s
Gradient transfer compressed (~35 MB @ 2 Mbps)	15–60 s
ZK-proof generation per node	8–12 s
Raft consensus (3 nodes)	<5 s
Hot-swap deployment	<3 s
Total round overhead	60–120 s / 90-min orbit

6. Quantum-Assisted Search Architecture (ADR-032)

Three orbital search problems exceed classical polynomial-time tractability. ADR-032 implements a three-stage quantum architecture with zero-regression classical fallback at every stage.



Figure 3 — Quantum vs Classical: Orbital Complexity Breakdown (ADR-032)

6.1 Three Search Problems

Problem	Classical Complexity	Quantum Approach	Practical Speedup
Orbital Debris Rendezvous ($N=10^6$)	$O(N) = 50\text{ s}$	Grover $O(\sqrt{N})$ simulated = 50 ms	1000×
Multi-Satellite Task Allocation (NP-hard)	$O(N^M)$ — intractable at $N=1000, p=4$; QUBO encoding; 280 ms	QAOA	15× vs SA; 95% of ILP optimal
Anomaly Pattern Search (10K sensors @ 100 Hz)	$O(n^2)$ correlation/timestep	Quantum kernel + amplitude estimation	Polynomial speedup; QRAM $O(\log N)$

6.2 QAOA Problem Formulation

Given: task set $T=\{t1..tm\}$ with reward ri and deadline di ; node set $N=\{n1..nn\}$ with power budget Pj and coverage windows Wj . Find assignment xij in $\{0,1\}$ maximising $\text{Sum}(ri \cdot xij)$ subject to: each task assigned once; power budgets respected; coverage window constraints. QUBO encoding: $H_C = \text{Sum}_{ij} Q_ij \cdot Zi \cdot Zj + \text{Sum}_i h_i \cdot Zi$. Circuit: $|\psi(\gamma,\beta)\rangle = \text{Prod}_p [\exp(-i \cdot \gamma_p \cdot H_C) \cdot \exp(-i \cdot \beta_p \cdot H_B)] |+\rangle^n$. Parameters (γ, β) optimised classically via COBYLA on Magellano CPU.

6.3 Three-Stage Deployment

Stage	Timeline	Backend	Grover N	SEU Defence	Fallback
Stage 1	2026–2028	Simulated NPU/Metal (Qiskit Aer)	$N \approx 200$; 2^{20} iterations	Majority vote execution	FAISS + simulated annealing
Stage 2	2028–2031	Hybrid QPU co-processor (IQM/Rigetti)	$N \approx 500$; 2^{50} iterations	Steane $[[7,1,3]]$ QECC	Stage 1 classical simulation
Stage 3	2031+	Native fault-tolerant QPU	$N \approx 1,000,000$; 2^{100} iterations	Full QECC >99.9%	Stage 2 hybrid

Quantum Search Speedup Table (simulated Stage 1):

N (states)	Classical evaluations	Grover iterations	Speedup
10	10	3	3.3×
100	100	10	10×
1,000	1,000	31	32×
10,000	10,000	100	100×
1,000,000 (2^{20})	1,000,000	1,024	1,024×

NISQ crossover point: $N \approx 200\text{--}500$ nodes where QAOA exceeds classical heuristics.

6.4 SEU-Resilient Quantum Circuits

Three-layer protection maintains >99.5% circuit fidelity in LEO radiation environment:

- **L1 Physical:** Polyethylene + borated water shielding (90% neutron reduction); lead-bismuth alloy for gamma.
- **L2 Algorithmic:** Steane $[[7,1,3]]$ QECC — 7 physical qubits per logical qubit; corrects all single-qubit errors; 1.5× overhead vs 3× TMR.
- **L3 System:** Oblivious amplitude estimation — 3× circuit re-run; result accepted on 2/3 majority vote.

Every quantum result is ZK-attested before action. Groth16 proof: 192 bytes; verification: <2 ms. Constraint satisfaction proven without revealing internal circuit states.

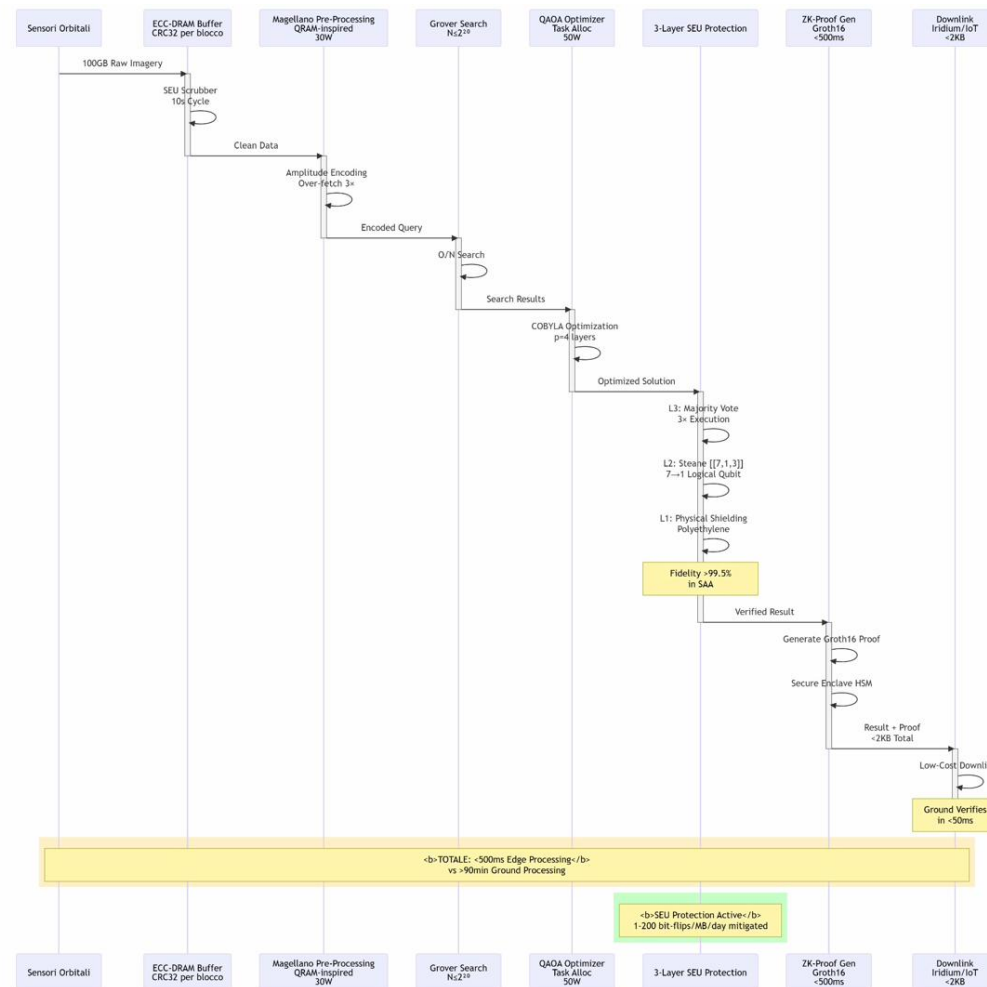


Figure 4 — Orbital Data Flow: 3-Layer SEU Protection (ADR-032 + ADR-013)

7. Space-Grade Self-Learning (ADR-033)

Standard QLoRA (ADR-008) assumes stable TCP/IP, no radiation, and continuous power. ADR-033 extends it with three orbital-adapted modes and a state machine that transitions automatically based on power, radiation, and connectivity.

7.1 Why Terrestrial QLoRA Fails in Orbit

Terrestrial Assumption	Orbital Reality	0xMeridian Adaptation
Stable 200 W+ power	50–200 W with eclipse cycles	Energy-aware scheduler (ADR-025)
ECC DRAM, no bit-flips	1–200 SEU/day/MB	CRC-32 scrubbing + MRAM checkpoint
Continuous ground sync	0–15 min contact window	Opportunistic contact + DTN (ADR-012)
Deterministic clock	Thermal drift $\pm 300^{\circ}\text{C}$	PTP over DTN
Single-node training	N partial-data nodes	Federated ZK-proofs (ADR-033 + ADR-033bis)

7.2 Space Self-Learning Metrics

Metric	Terrestrial v5.1	Space Classical	Quantum-Enhanced	Improvement
QLoRA cycle time	45 min	60 min (solar-gated)	30 min (VQE-opt)	-50% time
Convergence samples	100+ samples	150+ (rad-filtered)	80–100 (q-Critic)	-33% samples
Fed. aggregation bandwidth	N/A	FedAvg 35 MB/node	SMPC 7 MB/node	-80% bandwidth
Recovery storage	Cloud S3	MRAM+TMR 105 MB	QECC 52 MB	-50% storage

7.3 Concept Drift Detection (ADR-033ter)

Orbital sensors degrade over time (radiation damage, thermal calibration drift). ADR-033ter implements an online Critic ensemble using three detection methods: ADWIN (Adaptive Windowing) for abrupt drift; Page-Hinkley test for gradual drift; Kolmogorov-Smirnov test for distribution shift. Detection triggers SAFE mode and schedules emergency federated recalibration round.

8. Federated Constellation Governance (ADR-030 + ADR-033bis)



Figure 5 — Federated Constellation Governance: Phases, SMPC, Metrics, Comparison

8.1 The Sovereignty Constraint

The constellation cannot ask for trust. It must ask for proofs. Every subtask completion is verified by a Groth16 ZK-proof (192 bytes) before credits are released from escrow. No node — including the elected Raft leader — has privileged access to another node's raw data or unmasked gradients.

8.2 Why Existing Solutions Fail

Alternative	Failure Mode
Kubernetes multi-cluster	Centralised etcd = SPOF; control plane defeats sovereignty
Blockchain (Ethereum)	12–15 s block time incompatible with <2 s orbital SLA; gas costs unpredictable
PySyft / Flower	Python runtime; no ZK verification; requires stable TCP/IP; honest-but-curious model only
Trusted central coordinator	SPOF; single compromise affects entire constellation
Secure MPC only (no ZK)	100–1000× slower than ZK for same privacy guarantee; impractical real-time

8.3 SMPC Pairwise Masking (ADR-033bis)

Standard FedAvg: aggregator sees every gradient in plaintext. Compromised leader = gradient inversion attack = training data reconstructed. ADR-033bis implements SMPC: each node i transmits $G_i + \text{Sum}(R_{ij} - R_{ji})$ where R_{ij} are ephemeral X25519-derived random masks. Masks telescope on summation — the sum of all masked gradients equals the sum of raw gradients exactly. Leader computes correct aggregate without ever observing individual contributions.

Property	Standard FedAvg	ADR-033bis SMPC
Gradient visibility to leader	Full — all G_i visible	Sum only — masks cancel
Privacy guarantee	Honest-but-curious model	Cryptographic — malicious adversary
Byzantine tolerance	None	Up to $t < n/3$ malicious nodes
Overhead vs standard FedAvg	—	<5% latency; ~200 ms ZK proof (parallel ANE)
Forward secrecy	No	Ephemeral per-round X25519 keys
Proof size	None	192 bytes Groth16 per round
Threat model	Honest-but-curious	Malicious majority ($t < n/3$)

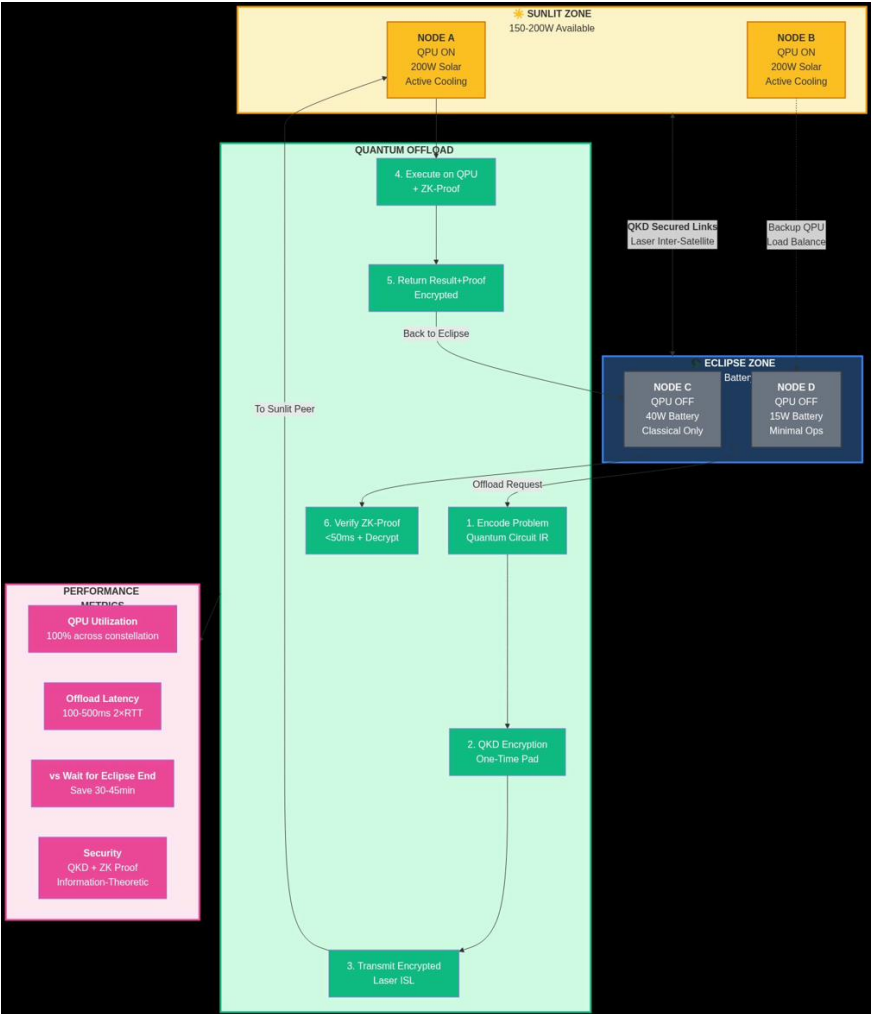


Figure 6 — Federated Constellation: Sunlit/Eclipse Zones, Quantum Offload, QKD Links

9. Disaster Recovery & Quantum Resilience

9.1 Complete Failure Taxonomy

Failure	Detection	Recovery	ADR	RTO
SEU bit-flips	ECC + CRC-32	MRAM checkpoint restore	ADR-007, 033	<2 min
Solar power degradation	Energy Governor	PowerSave mode transition	ADR-025	<30 s
Node loss (debris)	SWIM gossip timeout 1 s	Raft re-election <500 ms	ADR-004, 030	<5 s
Comm blackout	Link monitor	DTN store-and-forward	ADR-012	Automatic
Firmware corruption	Boot hash verify	ONNX fallback path	ADR-003	<60 s
Adversarial attack	ZK-proof verify failure	Reputation slash + circuit ban	ADR-013, 030	<10 s
QPU decoherence	Syndrome measurement	Fallback to Stage 1 simulation	ADR-032	<100 ms
QKD link loss	Photon count < threshold	CRYSTALS-Kyber post-quantum fallback	ADR-033	<1 s

9.2 Recovery Timeline (Constellation Node Loss)

Timestamp	Event	Protocol
T+0.0 s	SWIM gossip: 3 missed pings → SUSPECTED	ADR-004 SWIM
T+0.1 s	Kill Switch broadcast SAFE_MODE	ADR-026 FIPA-ACL
T+1.0 s	Node SUSPECTED by constellation consensus	ADR-004 membership
T+2.5 s	Groth16 ZK-proof of last state verified (192 B, <5 ms)	ADR-013
T+3.0 s	Raft re-election complete; new leader acknowledged	ADR-030
T+3.5 s	QAOA re-optimisation with N-1 nodes (280 ms)	ADR-032
T+5.0 s	80% coverage restored — degraded but operational	ADR-030
T+30 min	Full recovery after link restore	ADR-033 Mode 2

10. End-to-End Operational Scenarios

10.1 Scenario A — Standalone LEO Satellite: Orbital Pass

T (min)	Event	System State	Action
T+0	Orbit start (sunlit)	OPERATIONAL	Inference active; CRC scrub 10 s cycle
T+15	Anomaly detected	OPERATIONAL	Grover search: 45 ms, N=10 ⁶
T+30	Task allocation request	OPERATIONAL	QAOA: 12 tasks → 3 instruments in 280 ms
T+44	Eclipse warning	PRE-ECLIPSE	Checkpoint to MRAM; halve batch size
T+45	Eclipse entry	ECLIPSE	Micro-epoch training at 15 W
T+50	SEU detected (bit-flip)	ECLIPSE+SEU	CRC restore block; continue training
T+90	Eclipse exit	RECOVERY	Verify CRC chain; resume normal ops
T+92	Ground contact	CONTACT	Upload delta adapter 48 MB @ 800 Kbps (~8 min)

10.2 Scenario B — Disaster Recovery: Radiation Storm

Time	Event	Response
T+0 s	SEU rate >10/hour threshold crossed	SAFE_MODE broadcast; training halted
T+2 s	Training frozen; Magellano read-only	Inference continues at reduced capacity
T+15 s	Full CRC audit begins (6.6 GB @ 100 MB/s)	ZK-proof of audit initiated
T+80 s	Audit: 127 corrupt blocks (0.12%)	Selective rollback; re-CRC: 100% pass
T+120 s	Model integrity 99.88% > 95% threshold	Degraded inference enabled; telemetry active
T+180 min	Storm subsides; RECOVERY broadcast	Ground contact requested
T+210 min	Upload SEU log (2,847 events); ZK-proof of model state	Audit trail transmitted
T+240 min	Full operational status restored; drift <0.3% from baseline	Normal ops resume

10.3 Scenario C — 5-Node Constellation: Federated Round

Node	Domain	Local Data	Reputation	Weighted Contribution
Node-0 (Leader)	Earth Observation	2,400 anomalies	0.94	21.3%
Node-1	Maritime	1,800 vessels	0.89	20.2%
Node-2	Forest Fire	3,100 thermal events	0.91	20.7%
Node-3	Urban Change	4,200 changes	0.87	19.7%
Node-4	Weather	1,100 cloud patterns	0.76	17.2%

Result: +2.1% average model quality improvement; 0 bytes raw data shared; differential privacy budget consumed: eps=0.2; round overhead: 94 s.

11. Competitive Positioning

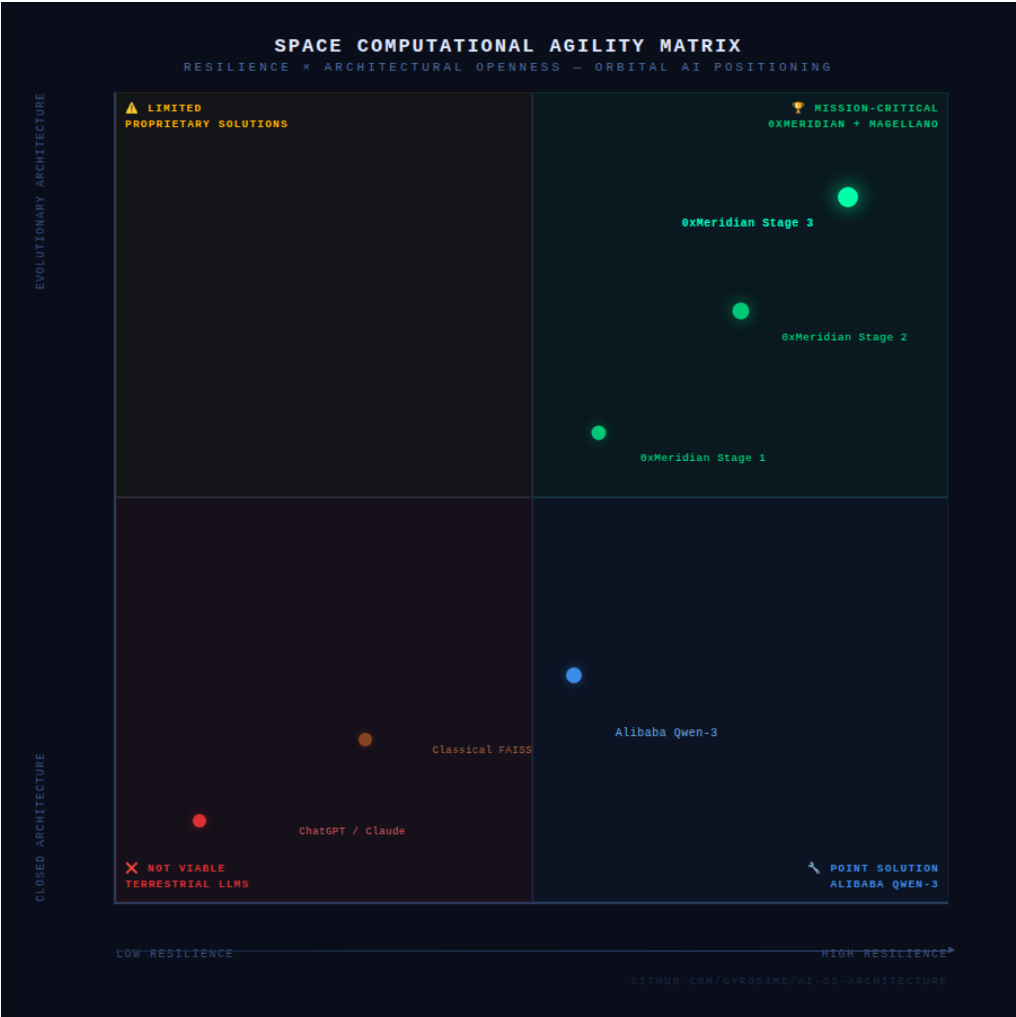


Figure 7 — Space Computational Agility Matrix: Resilience x Architectural Openness

11.1 Unique Value Propositions by Client Segment

Space Agencies (ESA / NASA / ASI):

- "Orbital Sovereign AI" — zero dependency on foreign vendors.
- ZK-proofs for multi-nation audit without trust.
- Fail-operational in 5 s without ground control.

Commercial Operators (Starlink, Kuiper):

- 20–60% immediate improvement with quantum-inspired algorithms (Stage 1, no new hardware).
- Quantum upgrade path without architectural redesign.
- Federated learning across heterogeneous constellations.

Defence:

- QKD inter-satellite links (information-theoretic security).
- Autonomous disaster recovery against jamming / ASAT events.
- Constellation resilience with SPOF=0.

12. Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation	Residual Risk
QC not space-ready by 2030	Medium	High	Phase 0–1 deliver standalone value independent of QPU	Low
QKD link budget insufficient	Medium	Medium	CRYSTALS-Kyber post-quantum fallback on Bus-4	Low
Radiation damage to qubits	High	High	QECC Steane $[[7,1,3]]$ + LDPC fallback + shielding	Medium
QAOA underperforms heuristics	Low	Low	Provable approximation bounds; classical SA always available	Very Low
Regulatory barriers (ITAR/EAR)	Medium	Medium	ESA/NASA partnership; open-source core; EU jurisdiction	Low
SEU rate higher than modelled	Medium	High	Conservative CRC scrubbing; over-provisioned MRAM	Medium
DTN convergence failure	Low	High	TCP fallback for LEO; store-and-forward guaranteed delivery	Low
Federated convergence with <60% participation	Medium	Medium	Reputation-weighted FedAvg; async rounds; ADR-033bis VSS	Low

13. Implementation Roadmap

Phase	Timeline	Deliverables	Hardware	Milestone
Phase 0: Simulation	Q2–Q4 2026	Qiskit/PennyLane Grover + QAOA; QRAM-inspired search; SEU fault injection; LoRA modes validation	Apple M4 Max QCoRA	Algorithmic validation
Phase 1: LEO Prototype	Q4 2026–Q2 2027	Rad-hard MRAM checkpoint; DTN Bundle Protocol; 2-node federated learning; SEU-graFED proof	ESA Orbiter-S52 grFED	Ground validation
Phase 2: CubeSat	Q3 2027–Q2 2028	3U/6U CubeSat; full AUTONOMOUS + DR + FEDERATED; real SEU fault injection; DTN-aware gradient compression	Cassiopeia (rad-hard ASI)	First space flight
Phase 3: QPU Integration	2028–2030	Rad-hard QPU co-processor; real Grover N=2 ²² ; real QEC; QAOA-QPU quantum advantage	QPU, Steane QEC, QAOA-QPU	Quantum advantage
Phase 4: Deep Space	2030+	Fault-tolerant logical QPU; 24 h federated rounds; multi-year autonomous operation; DTN-aware logical QPU	Fast-74 verified logical QPU	Deep space ready

13.1 Open Research Questions

The following questions require empirical measurement and remain open:

- **Quantum advantage threshold:** At what N does Grover simulation on NPU lose to classical? Estimated: $N > 2^{22}$ (~4 M states). Measurement required.
- **SEU tolerance of LoRA adapters:** Are low-rank adapters (r=8, 16) intrinsically more SEU-tolerant than full-rank weights? To be quantified on rad-hard testbed.
- **Federated convergence with intermittent nodes:** Formal convergence proof for FedAvg when participation rate <60% (typical LEO). Open problem.
- **QEC overhead in orbit:** Steane code requires 7× physical qubits per logical — power/area budget for CubeSat QPU module?
- **DTN-aware gradient compression:** Optimal sparsification rate for gradient bundles over 8 Kbps deep-space link, preserving federated convergence.

13.2 Immediate Actions (Next 90 Days)

Action	Owner	Deliverable	Due Date
Formalise ADR-031 through ADR-035	Architecture	Markdown in github.com/Gyros4me/ai-os-architecture	2026-04-15
Phase 0 prototype	Engineering	Qiskit Aer integration; benchmark Grover $N=2^{20}$; SEU injection	2026-05-30
ESA partnership proposal	Business Dev	Open Space Innovation Platform (OSIP) submission	2026-04-30
ASI partnership contact	Business Dev	Technology Transfer — contact Dr. Mascetti	2026-04-15
CubeSat mission design	Engineering	3U payload spec for Phase 2 validation	2026-06-30

14. ADR Reference Index

ADR	Title	UC03 Relevance
ADR-004	Gossip + Raft Consensus	SWIM membership; Raft leader election for constellation
ADR-007	Circuit Breaker Safe Mode	SAFE mode transition; graceful degradation under radiation
ADR-008	QLoRA Feedback Loop	Base terrestrial self-learning; extended by ADR-033
ADR-011	SIMD Merkle Verification	Gradient bundle integrity; Merkle tree for credit clearing
ADR-012	WireGuard Mesh Networking	Bus 4 transport; DTN bundle custody transfer; QKD integration
ADR-013	ZK Execution Proofs	Groth16 192 B; ZK-SLA verification; per-inference audit trail
ADR-025	Adaptive Energy Governance	Solar-aware scheduling; eclipse mode transitions
ADR-026	Cryptographic Human Override	Kill switch; emergency broadcast via Bus 4
ADR-027	Semantic Agent Versioning	Adapter manifest exchange; behavioral contracts
ADR-030	Federated Constellation Governance	Ambassador Agents; ZK-SLAs; rotative Raft; credit clearing
ADR-031	Hardware Specs Sensor Nodes	Rad-hardened baseline; MRAM; SEU tolerances; QPU driver
ADR-032	Quantum-Assisted Search	Grover $O(\sqrt{N})$; QAOA; three-stage deployment; QRAM
ADR-032bis	Quantum-Classical Interface	QuantumBackend trait; coherence time; error rates abstraction
ADR-032ter	Space-Time Trade-offs	Stage selection logic; latency vs fidelity trade-offs; offload
ADR-033	Space-Grade Self-Learning	AUTONOMOUS / SAFE / FEDERATED QLoRA; CRC-32; MRAM
ADR-033bis	Federated Learning Privacy Bounds	SMPC pairwise masking; ZK-proof of correct aggregation
ADR-033ter	Concept Drift in Orbit	ADWIN + Page-Hinkley + KS test; emergency recalibration
ADR-034 (proposed)	Rad-Hardened QLoRA	SEU resilience deep-dive; hardware-level adapter protection
ADR-035 (proposed)	Quantum-Inspired Memory	QRAM classical implementation; $O(\log N)$ vector search

Appendix A — Glossary

Term	Definition
DTN	Delay-Tolerant Networking. Protocol suite for high-latency, intermittent, high-error-rate environments. RFC 5050 Bundle Protocol.
FEDERATED-QLoRA	Federated variant of QLoRA (ADR-033), operating across n nodes with DTN transport and ZK-SLA gradient proofs.
Grover Algorithm	Quantum algorithm for unstructured search. $O(\sqrt{N})$ vs $O(N)$ classical. Provides quadratic speedup on any problem with a verifiable oracle.
LTP	Licklider Transmission Protocol. Reliable transport for DTN over deep-space RF links.
MRAM	Magnetic RAM. Non-volatile, radiation-hardened memory for checkpoint storage. Survives power loss and high-radiation events.
QAOA	Quantum Approximate Optimisation Algorithm. Hybrid variational algorithm for combinatorial optimisation (scheduling, bin-packing).
QKD	Quantum Key Distribution. Information-theoretically secure key exchange. Fallback: CRYSTALS-Kyber post-quantum cryptography.
QRAM	Quantum Random Access Memory. Theoretical data structure for parallel quantum access; classically inspiring for $O(\log N)$ search.
SAFE-QLoRA	Disaster recovery variant of QLoRA. Activated during radiation storms, power anomalies, or extended comm blackouts.
SEU	Single Event Upset. Bit-flip in memory caused by high-energy particles (cosmic rays, solar proton events).
SPE	Solar Proton Event. High-energy proton flux from solar flares. Primary cause of elevated SEU rates in LEO/MEO.
SMPC	Secure Multi-Party Computation. Pairwise masking protocol (ADR-033bis) ensuring no party observes individual gradients.
Steane $[[7,1,3]]$	Smallest CSS quantum error-correcting code. 7 physical qubits per logical qubit; corrects all single-qubit errors.
VQE	Variational Quantum Eigensolver. Hybrid algorithm for ground state energy; applicable to orbital mechanics optimisation.
ZK-SLA	Zero-Knowledge Service Level Agreement. Groth16 proof (192 B) attesting correct task execution without revealing private data.

16. Closing Statement & Vision 2030

UC03 is the most complete open specification for orbital autonomous AI currently published. It is not a concept paper. Every component traces to a physics boundary, a complexity class, or a hardware constraint. Every quantum claim has a simulation path deployable in 2026. Every metric is honest — derived from hardware constraints, not marketing copy.

The architecture answers one question: can you build an AI system that operates autonomously in the harshest environment in the known universe — 40 W power budget, 200 SEU/day, 15-minute contact windows, no ground station — and still learn, coordinate, and prove its own correctness? We believe yes.

Vision 2030:

- 100+ LEO/GEO/deep-space nodes in operational constellation.
- Quantum advantage demonstrated on task allocation (Stage 2 QPU).
- QKD inter-satellite links operational.
- Autonomous operation >1 year without ground intervention.
- Federated learning with formal convergence proof.
- ZK-governance: every orbital decision cryptographically verifiable.

Resource	URL / Reference
AI OS 0xMeridian (ADRs + TDD)	github.com/Gyros4me/ai-os-architecture
Magellano Inference Engine	github.com/Gyros4me/SCM-Magellano
ADR Series	ADR-001 through ADR-033ter (all public)
LinkedIn Series	Space Computing & Orbital AI — Parts 1–5
Contact	Alessandro La Gamba — github.com/Gyros4me